Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/29/2006
PI Name:	Nalette, Timothy Ph.D.		
Project Title:	Advanced CO2, Humidity, and Trace Contaminant Control System for Closed Loop Life Support Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	t.nalette@hs.utc.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	860-654-2403
Organization Name:	Hamilton Sundstrand Space Systems International, Inc.		
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PI Address 2:			
PI Web Page:			
City:	Windsor Locks	**State:**	CT
Zip Code:	06096-1010	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	08/11/2004	**End Date:**	09/30/2007
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ04HF73A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Conduct basic research associated with amine based sorbents for combined humidity and carbon dioxide control for closed loop life support. The basic research will investigate alternate sorbent formulations and configurations that will address enhanced kinetics and loadings to enhance overall system performance. These enhancements will result in reduced system mass, volume and power, while also enhancing overall vehicle integration issues associated with the addition of the humidity control function. Based on the results of the basic sorbent research, we will design, fabricate, assemble and test a full scale, deliverable prototype system.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	The following major tasks were completed during the period Aug 2004-Aug 2005. Completed survey of available alternate amine support materials. Define requirements of carbon supports to TDA. Obtained various samples of amine support materials to be evaluated. against baseline SA9T sorbent. Generated draft of system requirements. Supported a status review and presentation of status in Feb 2005 at JSC. Manufactured and tested numerous sorbents using carbons,silicas, polymeric supports as alternates to baseline PMMA- performance to date is only 60-65%of the baseline SA9T relative to equilibrium CO2 loadings. Manufactured and loaded with SA9T sorbent - a 4 chamber subscale test bed - delivered to MSFC for baseline testing and trace contaminant loading tests. MSFC has built test rig, run baseline test cases for rig check out and has completed the definition of analytical methods and test plan. University of Hartford has completed a subscale test rig to be used in the evaluation of new sorbents to be synthesised at both the Univ of Hartford and the Univ of CT. Both universities have defined synthesis routes for alternate amine based sorbents and generated samples of a number of new amines that will be tested in year 2 of this effort for CO2 /H2Ocapacity. The status of the year 1 effort was presented at ICES in Rome, Italy, ICES 2005-01-2865.
Bibliography Type:	Description: (Last Updated: 02/19/2009)
Papers from Meeting Proceedings	Nalette T, Reiss J, Filburn T, Mahan E, Seery T, Weiss B, Smith F, Perry J. "Development of an Amine-Based System for Combined Carbon Dioxide, Humidity, and Trace Contaminant Control." 35th International Conference on Environmental Systems, Rome Italy, July 2005. ICES paper 2005-01-2865, July 2005. , Jul-2005